

URBAN AND CRITICAL INFRASTRUCTURE RESILIENCE

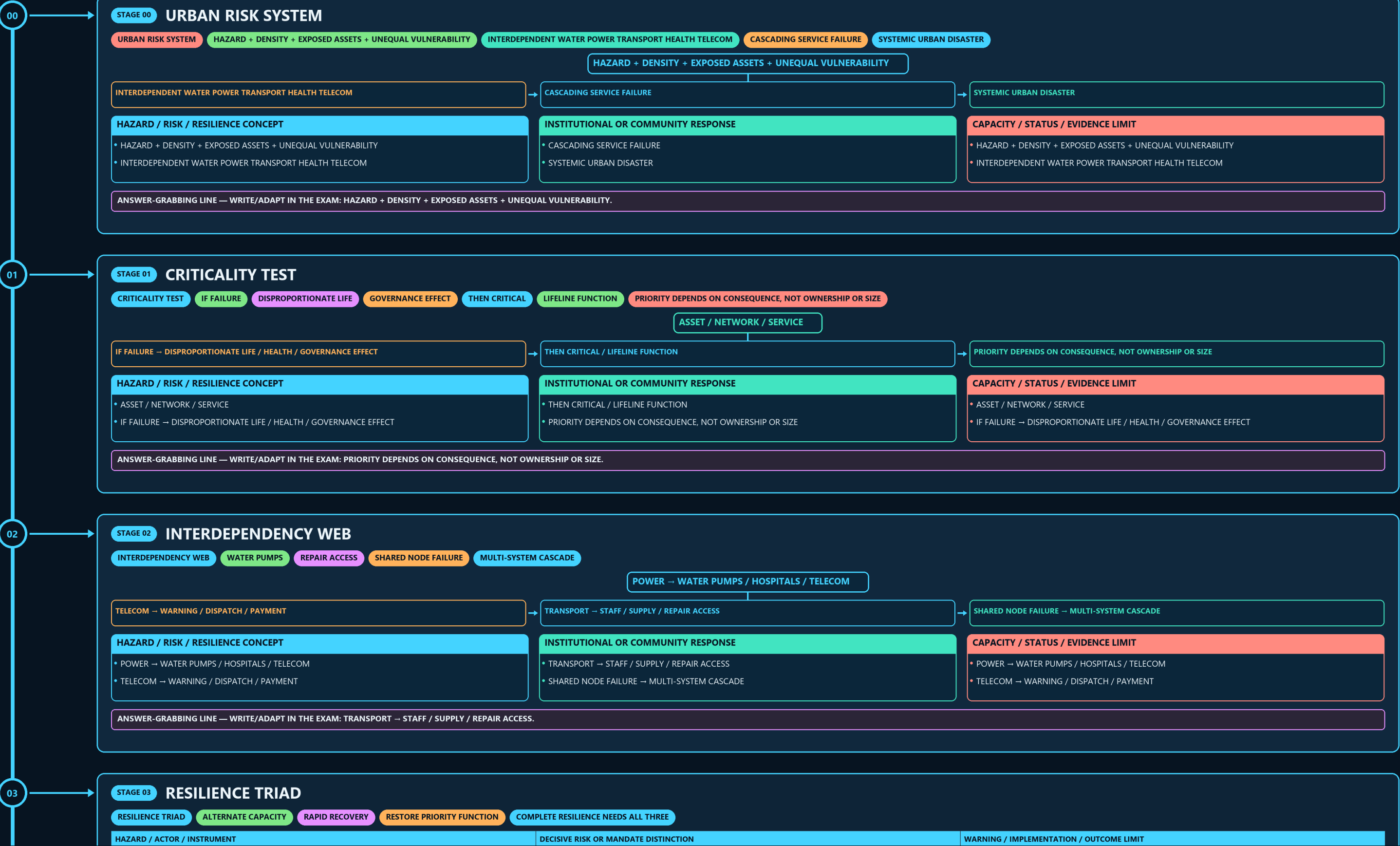
URBAN RISK SYSTEM → CRITICALITY TEST → INTERDEPENDENCY WEB → RESILIENCE TRIAD → CONTINUITY PLAN → URBAN PLANNING STACK

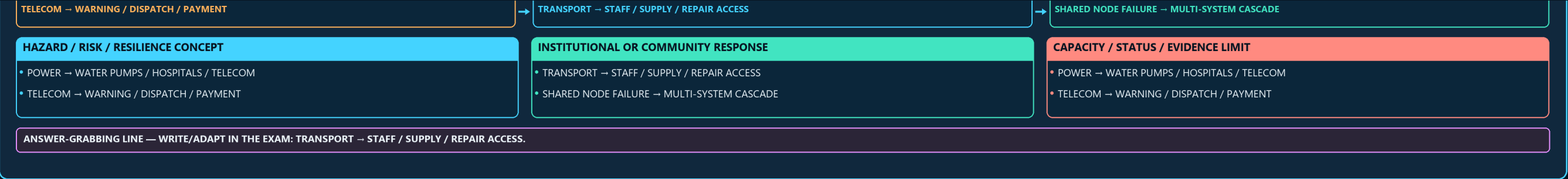
Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

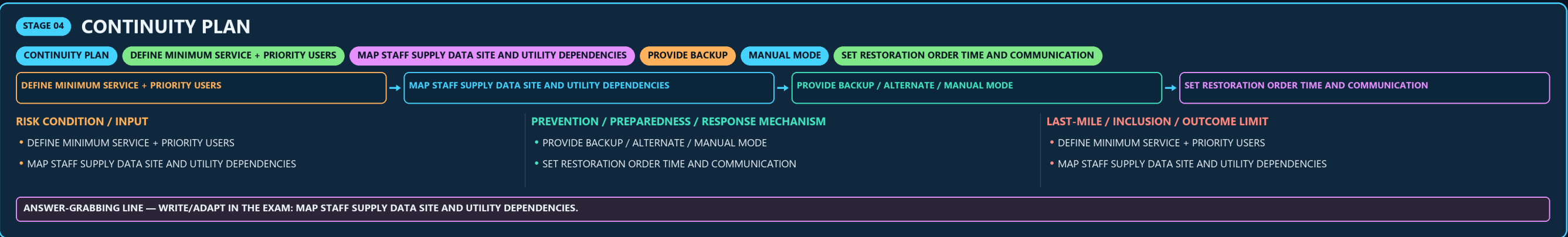




03



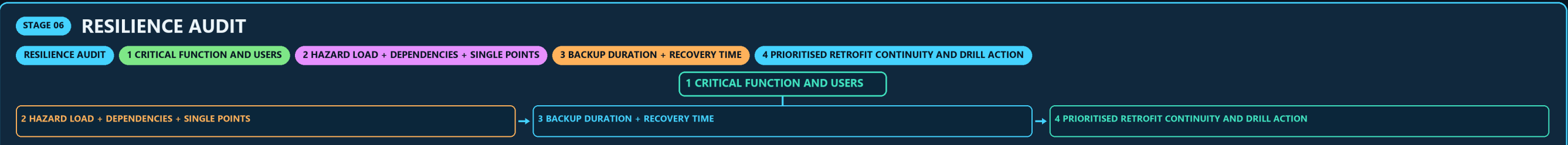
04



05



06



- LAND USE / FLOODPLAIN / DRAINAGE PATH
- BUILDING AND DEVELOPMENT CONTROL

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATER BODY DRAIN SOLID-WASTE AND MAINTENANCE; WARD-CITY-CATCHMENT COORDINATION.

1 CRITICAL FUNCTION AND USERS

→ 4 PRIORITISED RETROFIT CONTINUITY AND DRILL ACTION

LAST-MILE / INCLUSION / OUTCOME LIMIT

- 1 CRITICAL FUNCTION AND USERS
- 2 HAZARD LOAD + DEPENDENCIES + SINGLE POINTS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 2 HAZARD LOAD + DEPENDENCIES + SINGLE POINTS.

WARNING / IMPLEMENTATION / OUTCOME LIMIT
WARNING: The use of this tool is limited to the scope of the project. It is not intended for use in other contexts. IMPLEMENTATION: The tool is implemented in a way that ensures it is used correctly and safely. OUTCOME LIMIT: The tool is designed to achieve a specific outcome, and its use is limited to that outcome.

STRUCTURE / FIRE / FLOOD SAFETY

POWER WATER CLINICAL SUPPLY AND TELECOM

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- STRUCTURE / FIRE / FLOOD SAFETY
- POWER WATER CLINICAL SUPPLY AND TELECOM

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ACCESS STAFF REFERRAL AND SECURITY; CONTINUE CARE WHILE DAMAGE IS REPAIRED.

CAPACITY / STATUS / EVIDENCE LIMIT

- WHO LIVES IN HIGHEST EXPOSURE?
- WHO DEPENDS ON INTERRUPTIBLE OR INFORMAL SERVICES?

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WHO CAN REACH WARNING SHELTER HEALTH AND TRANSPORT?.

ULB / UTILITY / DEVELOPMENT AUTHORITY / DISTRICT

→ **RECOMMENDATION / ENABLING LAW / NOTIFICATION / OPERATION**

CAPACITY / STATUS / EVIDENCE LIMIT

- ULB / UTILITY / DEVELOPMENT AUTHORITY / DISTRICT
- SECTOR REGULATOR + SDMA / NDMA

09

STAGE 09 GOVERNANCE MAP

GOVERNANCE MAP

DEVELOPMENT AUTHORITY

SECTOR REGULATOR + SDMA

COORDINATION DOES NOT ERASE OWNER DUTY

ENABLING LAW

ULB / UTILITY / DEVELOPMENT AUTHORITY / DISTRICT

SECTOR REGULATOR + SDMA / NDMA

COORDINATION DOES NOT ERASE OWNER DUTY

RECOMMENDATION / ENABLING LAW / NOTIFICATION / OPERATION

HAZARD / RISK / RESILIENCE CONCEPT

- ULB / UTILITY / DEVELOPMENT AUTHORITY / DISTRICT
- SECTOR REGULATOR + SDMA / NDMA

INSTITUTIONAL OR COMMUNITY RESPONSE

- COORDINATION DOES NOT ERASE OWNER DUTY
- RECOMMENDATION / ENABLING LAW / NOTIFICATION / OPERATION

CAPACITY / STATUS / EVIDENCE LIMIT

- ULB / UTILITY / DEVELOPMENT AUTHORITY / DISTRICT
- SECTOR REGULATOR + SDMA / NDMA

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RECOMMENDATION / ENABLING LAW / NOTIFICATION / OPERATION.

10

STAGE 10 SMART RESILIENCE FIREWALL

SMART RESILIENCE FIREWALL

DIGITAL TWIN

COMMAND PLATFORM

PRIVACY + CYBERSECURITY + INTEROPERABILITY

OFFLINE FALLBACK + HUMAN ACCOUNTABILITY

EFFICIENCY WITHOUT BUFFER

BRITTLE SYSTEM

HAZARD / ACTOR / INSTRUMENT

SENSORS / DIGITAL TWIN / COMMAND PLATFORM → AWARENESS
PRIVACY + CYBERSECURITY + INTEROPERABILITY

DECISIVE RISK OR MANDATE DISTINCTION

OFFLINE FALLBACK + HUMAN ACCOUNTABILITY
EFFICIENCY WITHOUT BUFFER → BRITTLE SYSTEM

WARNING / IMPLEMENTATION / OUTCOME LIMIT

SENSORS / DIGITAL TWIN / COMMAND PLATFORM → AWARENESS
PRIVACY + CYBERSECURITY + INTEROPERABILITY

HAZARD / ACTOR / INSTRUMENT

- SENSORS / DIGITAL TWIN / COMMAND PLATFORM → AWARENESS
- PRIVACY + CYBERSECURITY + INTEROPERABILITY

DECISIVE RISK OR MANDATE DISTINCTION

- OFFLINE FALLBACK + HUMAN ACCOUNTABILITY
- EFFICIENCY WITHOUT BUFFER → BRITTLE SYSTEM

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- SENSORS / DIGITAL TWIN / COMMAND PLATFORM → AWARENESS
- PRIVACY + CYBERSECURITY + INTEROPERABILITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SENSORS / DIGITAL TWIN / COMMAND PLATFORM → AWARENESS.

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STAGE 11 URBAN RESILIENCE ANSWER SPINE

URBAN RESILIENCE ANSWER SPINE

DEFINE SYSTEMIC URBAN RISK AND CRITICAL LIFELINES

MAP CASCADES

ROBUSTNESS REDUNDANCY CONTINUITY

ADD LAND USE CODES DRAINAGE AUDIT EQUITY AND GOVERNANCE

USE CDRI

TECHNOLOGY WITH STATUS AND OUTCOME FIREWALLS

DEFINE SYSTEMIC URBAN RISK AND CRITICAL LIFELINES

MAP CASCADES → ROBUSTNESS REDUNDANCY CONTINUITY

ADD LAND USE CODES DRAINAGE AUDIT EQUITY AND GOVERNANCE

USE CDRI / TECHNOLOGY WITH STATUS AND OUTCOME FIREWALLS

HAZARD / RISK / RESILIENCE CONCEPT

- DEFINE SYSTEMIC URBAN RISK AND CRITICAL LIFELINES
- MAP CASCADES → ROBUSTNESS REDUNDANCY CONTINUITY

INSTITUTIONAL OR COMMUNITY RESPONSE

- ADD LAND USE CODES DRAINAGE AUDIT EQUITY AND GOVERNANCE
- USE CDRI / TECHNOLOGY WITH STATUS AND OUTCOME FIREWALLS

CAPACITY / STATUS / EVIDENCE LIMIT

- DEFINE SYSTEMIC URBAN RISK AND CRITICAL LIFELINES
- MAP CASCADES → ROBUSTNESS REDUNDANCY CONTINUITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: USE CDRI / TECHNOLOGY WITH STATUS AND OUTCOME FIREWALLS.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

CDRI'S PRECISE INSTITUTIONAL PURPOSE

TO DEVELOP "COMMON

URBAN DISASTER MANAGEMENT AUTHORITY STATUTORY STATUS

FIVE-CITY URBAN-FLOOD EVIDENCE BASE AS A CRITICAL-INFRASTRUCTURE

CDRI'S CATASTROPHE-BOND AND RESERVE-FUND TOOLS AS UNDERUSED IN

CRITICALITY MAPPING IS IMPLIED BUT NOT NAMED AS A

LIFELINE-FAILURE IMPACT IS NOT EVENLY DISTRIBUTED

OPTIONAL RISK / RESILIENCE DEPTH

- CDRI's precise institutional purpose : to develop "common
- Urban Disaster Management Authority statutory status : the
- Five-city urban-flood evidence base as a critical-infrastructure
- CAUTION: CDRI's catastrophe-bond and reserve-fund tools as underused in

MANDATE, WARNING OR OUTCOME LIMIT

- CAUTION: Criticality mapping is implied but not named as a distinct
- CAUTION: Lifeline-failure impact is not evenly distributed: informal
- CAUTION: Retrofitting existing lifeline infrastructure versus new-build
- CAUTION: "Build Back Better" for management systems versus physical

COMPARATIVE HAZARD / GOVERNANCE NUANCE

- NOT: The "CDRI membership/Urban Flooding Cell exists, therefore urban
- CAUTION: International-standard-setting versus domestic-operational
- CDRI's financing thematic area names budget reserve funds and
- Any current claim about Indian cities' actual use of catastrophe

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.